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Reimagining instructional design: ChatGPT, Bloom's taxonomy, and the role of teacher expertise

Belal Abu Eideh *An-Najah National University, Palestine*

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ABSTRACT

This study explores the potential of ChatGPT, a generative artificial intelligence tool, to assist in designing instructional units based on Bloom's Taxonomy of cognitive objectives. It specifically investigates how teacher experience influences the quality of AI-generated content and examines the extent to which these units represent varying cognitive levels. Employing a qualitative, multiple case study design, the research involved three physics teachers—novice, intermediate, and expert—who independently used ChatGPT to develop a unit using objectives from the Palestinian 10th-grade physics curriculum. Data were collected through semi-structured interviews and a custom log documenting teacher-AI interaction. Analysis combined thematic analysis and deductive content analysis aligned with Bloom's Taxonomy. Findings revealed that experienced teachers generated more balanced units across cognitive levels, while the novice teacher's unit focused mainly on lower-order thinking skills. Five key themes emerged: initial enthusiasm, the need for repeated prompts, doubts about quality, AI dominance in dialogue, and overall satisfaction. The study concludes that while ChatGPT can support instructional design, it requires skillful guidance and pedagogical awareness. It recommends teacher training in AI use, policy development for curricular integration, and further research comparing AI-generated and official curricular content through cognitive frameworks like Bloom's Taxonomy.

1. Introduction

The study of artificial intelligence has made tremendous strides recently, with many facets of life being affected by this extraordinary breakthrough. Generative AI represents a groundbreaking advancement, distinguished by its powerful creative functionalities and extensive linguistic capabilities (Kayabaş, 2024). People's scientific and productive capacities have been significantly impacted by the extensive availability and frequent use of artificial intelligence programs.

Automated procedures powered by artificial intelligence have enabled the completion of complex tasks and the generation of outputs once considered unattainable, owing to their capacity to automate processes, analyze vast datasets, and provide predictive insights (Ruiz-Rojas et al., 2023). Beyond their useful features, these apps are designed to generate context within a conversation and user feedback may later inform the training of future models, but they do not continuously learn from interaction in real time (Rahman & Watanobe, 2023).

The rapid advancement of generative artificial intelligence has placed education at the forefront of its influence, positioning it as one of the most affected domains and reshaping its practices and paradigms

(Grassini, 2023). One of the most notable influences has come from the ChatGPT application, which has asserted itself as a cognitive partner—contributing to the reshaping of the teacher–learner relationship and influencing educators' perspectives on knowledge, learning, assessment, and other educational concepts (Lo, 2023). Educators are now recognizing how much these tools are shaping the classroom and are considering how to use them in meaningful ways without losing sight of core educational values (Choi et al., 2024). One of the biggest changes has been the role of artificial intelligence in shaping learning materials. With these technologies, coursework can be created more quickly, made more diverse and interactive, and tailored more closely to students' needs, ultimately raising the quality of learning experiences (Dickey & Bejarano, 2024).

Lesson plans, learning activities, test questions, and enrichment materials can now be written using tools like ChatGPT, which has been shown to aid teachers in their pedagogy by offering functionalities like lesson planning, activity construction, and test question generation (refer to Table 3 in Dickey & Bejarano, 2024). This shift has allowed teachers to dedicate more time to the emotional and counseling roles of teaching, rather than putting all their time towards habitual

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conventional activities (Meron & Araci, 2023).

Artificial intelligence tools now give educators a wide range of options they can adapt to their specific teaching needs, as shown in multiple studies. For example, Rasul et al. (2023) found that tools like ChatGPT allow teachers to shift their role from being the sole source of knowledge to fostering more collaborative and interactive learning. Similarly, AI-based applications provide both teachers and students with opportunities for self-directed learning by enabling them to ask questions and receive instant feedback, which helps build independence in developing skills (Bettayeb et al., 2024).

Overall, generative AI represents a significant shift in how teaching and learning are approached—shaping lesson planning, instructional design, and classroom practice. The move away from traditional methods toward more dynamic, collaborative learning environments signals a deeper transformation that goes beyond technology. It touches on the philosophical, methodological, and ethical dimensions of education. For this reason, educators and scholars must approach these changes thoughtfully, ensuring AI is used in ways that enhance the quality and impact of teaching and learning.

1.1. Research gap

There is an increasing need for research that examines how generative AI tools like ChatGPT are used in education. This requires both technical analysis, focusing on how the technology generates text, and qualitative studies that look at their impact on teaching and learning in real-world contexts. Several studies have indicated that ChatGPT is a promising tool for supporting educational planning and content design (Li, 2024; Utami et al., 2025). However, these studies have primarily focused either on analyzing teachers' perceptions and attitudes toward the use of generative AI applications or on reviewing their technical features from a general standpoint, without delving into the structural analysis of their educational outputs based on objective frameworks."

Although some experimental studies have attempted to explore the extent to which ChatGPT can generate acceptable educational content, few have dedicated their objectives to examining the alignment of such content with the educational goals established by specialized committees, which form the foundation of formal curricula (Ullmann et al., 2024).

Moreover, several studies in the educational literature suggest that instructional objectives considered the central structural component of materials designed using ChatGPT have not received sufficient analytical attention, and only limited systematic analyses have examined their alignment with the intended learning outcomes (Doyle et al., 2025; Ullmann et al., 2024).

Accordingly, this study aims to address a research gap related to the limited number of studies that analyze the actual design outputs generated by the ChatGPT when used by real teachers in a school setting, based on specific inputs, yet without constraints on the nature of the output.

Also, If the design outcomes generated through ChatGPT are to be considered as co-constructed products between the tool and the user, then understanding the nature of the interaction between the teacher and generative artificial intelligence applications becomes essential for interpreting these outputs. ChatGPT does not operate in a vacuum; rather, it responds to the user's prompts, shaping its outputs based on the user's wording, revisions, and persistence in requests. This emphasizes how crucial it is to consider the teacher's expertise as a factor that affects the interactional behavior with ChatGPT, the knowledge of its potential and limitations, and the tactics employed to take advantage of it, rather than just as a personal variable (Zamfirescu-Pereira et al., 2023). Accordingly, even when employing the same tool and the same input, disparities in the participating teachers' levels of expertise may produce different educational outcomes (Lee et al., 2024). Despite some early findings in recent studies indicating variations in output quality based on users' educational backgrounds and levels of proficiency, this

dimension has not received enough attention in the literature to date.

1.2. Purpose of the study

This study aims to conduct a comparative analysis of three instructional units designed by teachers with varying levels of experience using ChatGPT. The goal is to understand how these educators utilize the tool in designing instructional units and how their differing levels of expertise influence the formulation of learning objectives in the context of a unit design.

Additionally, the study seeks to compare these AI-generated units with a parallel unit extracted from the formal curriculum, developed for the same educational level, in order to identify similarities and differences between the designed and the formal units. This comparison will help assess the extent to which generative artificial intelligence adheres to established learning objectives when interacting with teachers who vary in their levels of experience during the design process.

The study also aims to interpret the patterns of interaction that occur between teachers and ChatGPT and to explore how this interaction affects both the educators and the resulting instructional content.

1.3. Research contribution

This study seeks to address a practical gap in previous research by proposing a qualitative analytical model that compares human-generated outputs with those produced by generative artificial intelligence using ChatGPT. Specifically, it compares an instructional unit derived from a formal curriculum with units designed using ChatGPT to achieve the same educational objectives. The study is distinguished by its focus on the real experiences of three teachers who used the same tool to design the same unit, allowing for the analysis of differences in instructional design outcomes resulting from variations in expertise and educational perspectives, despite the consistency of the tool and subject matter.

Moreover, the study offers a methodological contribution by integrating both descriptive and inferential analyses of instructional units generated through ChatGPT, alongside thematic analysis of teacher interviews. This approach provides a deeper interpretive insight into how teachers interact with AI tools in instructional design and offers practical indicators for leveraging these tools to create more integrated and contextually relevant educational units.

1.4. Research questions

This study seeks to answer the following questions:

1. What is the distribution of the educational objectives included in the units designed using ChatGPT according to Bloom's cognitive taxonomy?
2. What are the similarities and differences between the learning objectives in the units designed using ChatGPT and those in the official unit, in terms of cognitive depth according to Bloom's Taxonomy?
3. How did the teachers describe their experience using ChatGPT to design an instructional unit in terms of their interaction with the tool, the challenges they encountered, and their evaluation of the resulting outputs?

2. Literature review

2.1. The transformative role of generative AI in education

The increasing use of generative artificial intelligence tools has led to a fundamental shift in our understanding and perception of the role of technology in education. Previously regarded as a supportive means, it is now - under this growing usage - considered an active partner in the design of learning itself. This shift has also influenced current

educational research and may further lead to a redefinition of the traditional relationships between the teacher, the learner, and the content, especially in the context of the rapid development of large language models such as ChatGPT.

Many studies have contributed to documenting this structural transformation, prompting educational researchers to tackle the central question of whether generative AI tools can actually play the role of the “instructional designer.”

In this context, research frameworks such as the GAIDE model developed by [Dickey and Bejarano \(2024\)](#) emerge as structures to conceptualize the role of generative AI in educational content design, starting from goal generation, passing through content formulation, and culminating in refinement and thorough review processes. This framework, from an interactive perspective, restores to the teacher the authority that had been lost due to the intensive use of generative AI applications - granting them pedagogical oversight -while keeping generative AI as a stimulus for thinking rather than a replacement for pedagogical reflection.

This vision finds clear resonance in the CourseGPT experience attempted by ([Nazar et al., 2024](#)), which offers a detailed conception of how AI can be employed to generate a full university course, tailored and aligned with lecture files, course requirements, and the institutional context.

In the era of open internet access, AI can no longer remain confined to institutional use. Rather, it expands to include individual initiatives by both teachers and students, which may lead to educational institutions losing control over pedagogical aspects. To this end, [Ullmann et al. \(2024\)](#) conducted a study in an open distance learning environment where ChatGPT was used to design learning units, generate related summaries, create assessments aligned with those units' goals, and transform learning outcomes into interactive activities without compromising the institution's official pedagogical framework. Such studies point to the emergence of an “assistive pedagogical intelligence,” operating in the margins without making learners feel the absence of the human role.

2.2. Benefits, risks, and institutional practices

Despite all the capabilities that generative AI applications offer in planning and designing educational content, educational studies have not overlooked the risks associated with unregulated, excessive use of these tools. [Rahman and Watanobe \(2023\)](#) observed that uncontrolled, random use inevitably leads to overreliance on generative AI tools by learners, which may negatively impact the effectiveness of higher-order cognitive processes and weaken self-assessment ability which is a central value in Bloom's cognitive domain framework.

This aligns with the findings of [Panagopoulos et al. \(2023\)](#), whose study revealed that many of ChatGPT's educational outputs lack precise cognitive analysis, and that it is the context that determines whether the use of AI stimulates thinking—or hinders it.

Conversely, some institutions have begun adopting methodologies that allow them to steer the use of generative artificial intelligence toward pedagogically coherent practices. For example, [Ruiz-Rojas et al. \(2023\)](#) implemented the 4PADAFE instructional design matrix to integrate generative AI tools into the development of digital learning environments. The study found that AI does not replace instructional designers but instead supports them by prompting deeper reflection on pedagogy and helping clarify learning objectives and assessment activities. [Ullmann et al. \(2024\)](#) reported that while the materials produced in course development workshops with AI were well-structured, they still required careful review and alignment with learning goals. This suggests that creating high-quality educational materials depends not on AI alone, but on the expertise of educators who ensure the content meets academic standards.

2.3. Cognitive depth and Bloom's taxonomy

Several studies have begun to question how educators and facilitators can guide AI-generated material so that it remains cognitively coherent, particularly in relation to Bloom's Taxonomy. However, current literature shows that little systematic work has been done to compare AI-generated content with established curricula or to analyze it using Bloom's framework. This gap points to a deeper issue: a limited understanding of the intellectual processes needed for strong instructional design and the absence of standardized tools for evaluating AI-produced materials. Key questions remain: Does AI-generated content follow a deliberate cognitive sequence? And does it align with the hierarchical levels outlined in Bloom's Taxonomy? Can it meet the standards of depth and coherence expected in well-designed educational units? These questions remain unanswered, and addressing them provides a clear rationale for the present study.

Actually, the logic and structural coherence - described by [Ullmann et al. \(2024\)](#) as characteristics of AI-generated content - stem from the fact that generative models are trained on linguistic patterns extracted from vast quantities of educational documents. However, this statistical repetition does not necessarily mean that the content produced by AI genuinely adheres to the cognitive sequencing that characterizes well-designed instructional units. Thus, it becomes imperative to develop analytical tools that examine the deep structure of the content, explain its conceptual levels, and serve as indicators of quality in educational AI applications.

Some studies have approached this major challenge, but none have addressed it directly. [Rahman and Watanobe \(2023\)](#) identified several challenges in AI-generated content, such as conceptual ambiguity, repetition, and difficulty handling abstract ideas. These issues point to a lack of structural coherence in the material produced. Similarly, [Apata et al. \(2024\)](#) found that when trainees were taught to use ChatGPT, they mostly treated it as a smart assistant rather than using it for deeper inquiry or critical analysis of the content. This suggests that ChatGPT is being used more as a support tool than as a resource for advanced pedagogical work.

When compared with traditional instructional design, the literature shows that AI-generated educational materials often lack focus on hierarchical organization or cognitive development. For instance, if a teacher asks ChatGPT to generate a lesson on plant reproduction, the tool can provide definitions, examples, and simple exercises. However, the output usually remains at the level of comprehension and application, even when the user specifically requests more complex tasks or analytical questions. This confirms the findings of [Ullmann et al. \(2024\)](#): Instead of creating new pedagogical logic, generative AI likes to reimplement existing patterns.

It is helpful here to distinguish between two types of AI-generated content: real-time interactive content (such as instant answers to student queries) and long-form structural content (such as full instructional units or lesson plans). While the former has received broad coverage in educational literature, the latter remains largely absent from cognitive-level analysis. The absence of studies analyzing content generated by systems like CourseGPT, AutoGPT, or 4PADAFE through the lens of classification models such as Bloom's Taxonomy reflects not only a gap in the literature but also a gap in the design logic underpinning these systems.

Practically speaking, many studies have explored the first type of content, but the second type—which concerns structured curricular outputs—continues to be underrepresented in educational scholarship.

Using Bloom's Taxonomy to analyze AI-generated content should not be treated as a mere classification exercise; rather, it should serve as a framework for understanding the type of “thinking” embedded in the generated text. Does the text require learners to recall, understand, apply, analyze, synthesize, or evaluate? Or does it remain confined to the lower levels of cognition? More concerningly, does it sometimes produce seemingly analytical texts that are, at their core, rewordings of

low-level thinking? This points to the need for two levels of analysis when reviewing AI-generated content: a surface-level reading of the language and a deeper cognitive evaluation. Current studies often miss this second layer. [Nazar et al. \(2024\)](#) highlight the issue further, showing that while language models follow user instructions effectively, they often over-align with form and produce content that looks polished but lacks depth. As a result, users must continually monitor and evaluate AI outputs to avoid pedagogical superficiality.

The problem is not only with the technology itself but also with how it is applied in educational settings. These tools are often used by students and teachers who lack formal training in instructional design, which leads to shallow or formulaic learning materials. [Koenig \(2024\)](#) noted that many teachers struggle to adapt their pedagogical content into AI systems, especially without professional development that addresses the cognitive and ethical challenges of using such tools. Without this foundation, AI remains a text generator rather than an instructional designer capable of supporting sound educational principles.

[Gousopoulos \(2024\)](#) made a similar observation in their study of GPT-4o mini. While the tool showed promise for creating educational videos and narrative texts, it was far less effective at designing assessment activities aligned with specific learning outcomes. The researchers blamed this shortage on the model's failure to understand cognitive development since it relies on statistically probable linguistic patterns rather than conceptual depth. It must thus be explicitly directed by an instructional designer who has experience with frameworks like Bloom's Taxonomy.

On the other hand, there is research demonstrating how AI can be trained within a rigorous pedagogical logic. For instance, [VanDriel and Berry \(2012\)](#) suggested that teachers can develop "pedagogical content knowledge" concerning how to design questions and tasks according to Bloom's levels.

If such knowledge is translated into clear instructions for the AI, the output can approximate the desired cognitive standards. However, this process is not automatic; it requires the integration of AI tools into teacher preparation programs, as noted in the studies by [Apatá et al. \(2024\)](#) and [Utami et al. \(2025\)](#), where usage remains largely experimental and lacks institutional structure.

What is particularly striking in this context is that many of the studies celebrating the role of AI in education—such as [Castillo et al. \(2024\)](#) and [Kayabaş \(2024\)](#)—have focused on technical or psychological dimensions of usage (e.g., accessibility, or reduced learner anxiety), but have not delved into the cognitive depth of the generated content or its alignment with established educational models. This underscores the fact that evaluating the quality of AI-generated instructional units is not merely a matter of linguistic or technical refinement, but fundamentally a matter of cognitive integrity.

2.4. Identified gaps in the literature

Given these considerations, the present literature review assumes a foundational role, as it highlights a dual gap: first, the absence of studies analyzing AI-designed content through Bloom's Taxonomy; and second, the lack of comparative studies between such content and that drawn from official school curricula. This comparison is essential, since AI does not operate in a vacuum—it is increasingly positioned as a tool capable of replacing or at least matching human instructional design efforts. Without a rigorous cognitive evaluation of such claims, we risk endorsing tools that may produce educational content falling short of basic conceptual quality.

Despite all the capabilities that generative AI applications offer in planning and designing educational content, educational studies have not overlooked the risks associated with unregulated, excessive use of these tools. [Rahman and Watanobe \(2023\)](#) observed that uncontrolled, random use inevitably leads to overreliance on generative AI tools by learners, which may negatively impact the effectiveness of higher-order cognitive processes and weaken self-assessment ability, which is a

central value in Bloom's cognitive domain framework.

The lack of systematic analysis of educational content designed using artificial intelligence—whether in light of Bloom's Taxonomy or other educational frameworks—cannot be explained solely as a knowledge gap. Rather, it reflects a distortion driven by the structure of the technology market, which often promotes the use of such tools through the lens of speed and efficiency, while neglecting deeper pedagogical considerations. In this context, the educational literature documents several ways in which ChatGPT has been employed for content design or simplification, such as the study by [Liu et al. \(2024\)](#) on text simplification tools and the AutoGPT experiment in providing precise educational feedback ([Demirbek & Ekizer, 2024](#)).

However, only a limited number of these studies consider artificial intelligence as a generator of knowledge that should be critically examined through the lens of established learning theories.

In summary, the logical flow and structural coherence often seen in AI-generated content—described by [Ullmann et al. \(2024\)](#)—stem from the fact that generative models learn from the linguistic patterns found in large collections of instructional materials. However, this statistical patterning does not guarantee that the content follows the cognitive sequence expected in well-designed instructional units. This highlights the need for tools that can analyze the deeper structure of AI-generated content, break down its conceptual levels, and assess its pedagogical quality.

Some studies have touched on this issue but have not addressed it directly. For example, [Hasanein and Sobaih \(2023\)](#) identified problems in AI-generated content such as conceptual ambiguity, redundancy, and difficulty handling abstract concepts—early signs of weak structural coherence. Similarly, [Apatá et al. \(2024\)](#) found that when pre-service teachers were trained to use ChatGPT, they tended to treat it as a smart assistant rather than using it for questioning or critical analysis. This reinforces the idea that ChatGPT is often used as a support tool rather than for deeper pedagogical engagement.

2.5. ChatGPT in K-12 education

Recent systematic reviews show both the opportunities and risks of using ChatGPT in K–12 education. [Zhang and Tur \(2023\)](#) point out that ChatGPT can support teachers with curriculum design, lesson planning, and instructional material creation, while also enabling more personalized and flexible learning. These strengths highlight its efficiency and adaptability in classrooms.

However, concerns remain. Studies note risks around the accuracy and reliability of ChatGPT's outputs, as well as academic integrity issues such as plagiarism and overreliance ([Zhang & Tur, 2023](#); [Gödde et al., 2023](#)). Ethical risks are also significant, including misuse and the potential weakening of students' critical thinking if the tool is adopted uncritically ([Gödde et al., 2023](#); [Lo et al., 2024](#)). At the same time, researchers also see potential for innovative teaching methods and more engaging, technology-driven learning spaces ([Lo et al., 2024](#)).

[Lo et al. \(2024\)](#), after reviewing 72 empirical studies, found that ChatGPT influences all three dimensions of student engagement. Behaviorally, it can increase participation and task completion. Emotionally, students reported feeling entertained, satisfied, and happy, though sometimes also anxious or disappointed. Cognitively, it supported comprehension and confidence but raised concerns about reduced critical thinking and dependence on AI.

One practical example comes from [Bitzenbauer \(2023\)](#), who integrated ChatGPT into a high school physics class. In a pilot with 53 students, ChatGPT generated explanations on photons and wave-particle duality. Students then reviewed, edited, and compared these outputs with professional sources, along with generating and critiquing conceptual survey questions. These activities improved conceptual understanding while also encouraging critical evaluation of AI-generated content rather than passive acceptance.

Taken together, the literature suggests that ChatGPT has strong

potential to enhance learning and teaching in K–12 contexts. It can boost motivation, creativity, and subject-specific learning, such as in physics. Still, these benefits depend heavily on thoughtful pedagogy and ethical safeguards to address the risks of inaccuracy, overreliance, and weakened critical thinking.

2.6. The TPACK framework and Teacher–AI interaction

The TPACK model (Mishra & Koehler, 2006, 2008) points out that effective technology teaching is dependent on how teachers' technological knowledge (TK), pedagogical knowledge (PK), and content knowledge (CK) are fused. Recent research confirms that technological knowledge contributes greatly to how educators utilize AI-based tools such as ChatGPT. For instance, Tan et al. (2025) depicted that educators' AI proficiency positively predicted their TPACK levels, which in turn predicted elevated teaching performance. Similarly, Ning et al. (2024) determined that AI-related technological knowledge (AI-TK) exerted more pronounced influence on teachers' aggregate AI-TPACK than pedagogical or content knowledge alone, affirming the foundational role TK has in enforcing effective AI integration. Yue et al. (2024) also cited that K–12 teachers with greater technological knowledge were more capable of integrating AI tools meaningfully with pedagogy and content, whereas those with lower TK used AI superficially. In the same context, Ismaniati, Khairaty, Yusuf, and Supramono (2025) came to show that those teachers possessing a higher TPACK are more confident in using AI, meaning self-belief in using tools like ChatGPT depends directly on how much technology knowledge they possess. Together, these studies reveal that the quality of teacher–AI interaction is contingent not only on pedagogical intent but also on the technology proficiency of teachers, and therefore TK plays a deciding role in enriching learning experiences cognitively.

3. Research methodology

3.1. Research design

This study adopts a qualitative research methodology to understand how teachers utilize generative artificial intelligence tools—specifically ChatGPT—in designing instructional units, and to analyze the points of convergence and divergence between these AI-generated units and an instructional unit derived from the formal curriculum. The study relied on three data collection instruments: (1) content analysis of the instructional units designed using ChatGPT, (2) semi-structured interviews with the participating teachers, and (3) a tool for transcribing interactions between the teachers and ChatGPT.

This research is based on a multiple case study design, which is considered appropriate when the goal is to explore differences and contextual variations of a particular experience across different settings. Each of the three participating teachers was considered an independent case. Each teacher independently designed an instructional unit using ChatGPT, without intervention from the researcher. Each case was then analyzed separately, followed by a cross-case comparison of the three units. Subsequently, the AI-generated units were compared to a reference instructional unit derived from the official curriculum that addresses the same academic topic and targets the same student group for whom the teachers designed their ChatGPT-based units.

3.2. Study context

3.2.1. Participating teachers

Three public-school physics teachers from Nablus, Palestine were purposefully selected based on variation in years of experience, in order to explore their pedagogical differences in the use of ChatGPT in designing instructional units.

- The first teacher had only one year of teaching experience and represented the category of novice teachers, who are typically assumed to be more familiar and flexible with digital technologies.
- The second teacher had six years of experience and was considered to represent mid-career teachers who have accumulated pedagogical knowledge through practical practice.
- The third teacher had over twenty years of teaching experience and was classified as a highly experienced teacher. This group is more prone to what is pedagogically known as “professional burnout,” which may influence their interaction with new tools or their resistance to changing traditional instructional patterns.

The three teachers shared several academic and professional characteristics. All held bachelor's degrees in physics, graduated from the same university, and were teaching in public schools at the same grade level. Notably, none of them had received prior training in using ChatGPT, and their use of the tool was individual and non-intensive. Furthermore, none of them had previously participated in curriculum design teams affiliated with the Palestinian Ministry of Education. This background suggests that the three teachers were not expert instructional designers with direct involvement in curriculum development. Instead, they represented natural field-based experiences of practitioners working in general education settings, which gives the study findings a realistic character.

All teachers were invited to use ChatGPT to construct a lesson unit from scratch, i.e., that within this study, it meant commencing the process from solely the listed learning objectives found in the official tenth-grade physics teacher's handbook (Palestinian Ministry of Education, 2019). No other context materials (e.g., course information or exemplar units) were provided. Teachers were required to adhere strictly to these formal objectives and to enter them into ChatGPT as their initial point before beginning to design. Other than this, there were no restrictions or pre-established instruction design models used: teachers were free to use ChatGPT in the manner they preferred, using methods and prompts of their own selection. This enabled the study to examine the role of ChatGPT as an instructional design assistant across varied professional contexts.

The participating teachers demonstrated clear intrinsic motivation to engage in the experience of using ChatGPT for instructional unit design. They expressed interest in exploring ChatGPT's potential for instructional planning.

3.2.2. The formal instructional unit

The unit on “Mechanics”—the first unit in the official tenth-grade physics curriculum (Palestinian Ministry of Education, 2018), was selected as the reference unit against which the AI-generated units would be compared. The unit is taught to tenth-grade students, typically aged 15, and consists of four lessons:

1. Physics and Measurement
2. Vectors
3. Describing Motion
4. Newton's Laws

The unit covers concepts such as measurement, vectors, and motion, which enable learners to interpret familiar phenomena in their local environment such as car movement, falling objects, and changes in position over time.

The Mechanics unit, extracted from the official textbook, is characterized by a semi-detailed design. It provides both teachers and students with specific guidance, including suggested activities, discussion points, and assessment questions. In physics tenth grade teachers' guide (Palestinian Ministry of Education, 2019), the unit is relatively simple in structure and includes approximately 141 learning objectives, distributed across Bloom's six cognitive levels (remembering, understanding, applying, analyzing, evaluating, and creating), as documented in the

teacher's guide analysis (Appendices 1). Table 1 shows the distribution of the Official Unit's Objectives According to Bloom's Taxonomy of Cognitive Objectives.

3.3. Semi-structured interviews

The study employed individual semi-structured interviews with the three teachers after they had completed designing their instructional units using ChatGPT. The aim was to understand both the overall instructional design experience and the nature of the interaction between the teachers and the application, as well as the decision-making strategies they employed during the process.

Each interview lasted between 30 and 40 min, was conducted individually, and-with the participants' consent-was audio recorded and later transcribed for analysis.

The interviews were guided by predefined thematic axes, which included:

1. Teachers' general perceptions of the experience
2. Nature of interaction with ChatGPT
3. Key challenges and difficulties faced during planning with ChatGPT
4. General evaluation of the content generated using ChatGPT

The interview transcripts were analyzed using thematic analysis, starting with open coding to identify recurring patterns of meaning, followed by organizing codes into subcategories and overarching themes that reflect the teachers' pedagogical thinking and design behaviors. The results of the interviews were then compared with the content analysis of the instructional units created by the same teachers to reinforce interpretation through data triangulation.

3.4. Interaction monitoring framework with ChatGPT

To accurately document the teacher's design decisions while working with ChatGPT, a specialized observation framework was developed to track the interaction process during the designing process. All interactions with ChatGPT were conducted in Arabic, reflecting the language of instruction in Palestinian schools. The version used during the data collection period was ChatGPT-4. This framework comprises five core components:

1. **Initiating the Interaction:** This section captures how the teacher began engaging with ChatGPT, including the types of questions or prompts used to start the conversation.
2. **Inputs Provided for Unit Design:** Here, the teacher lists the specific information supplied to the tool such as learning objectives, lesson count, lesson topics, session allocation per lesson, or any other relevant inputs used to guide the designing process.
3. **Guidance Given During the Design Phase:** This part records all instructional directives the teacher communicated to ChatGPT throughout the designing process.
4. **Challenges Encountered During Design:** In this section, the teacher reflects on any difficulties or obstacles faced while using the tool to design the unit.

5. **Overall Evaluation of the Generated Content:** The teacher provides a general assessment of the final outputs, expressing their level of satisfaction with the results of the design process.

3.5. Content analysis of the instructional units

The three instructional units designed by the participating teachers using ChatGPT were subjected to both descriptive and interpretive analysis in order to explore the pedagogical features embedded within these units. This analysis was conducted in alignment with the study's research questions and thus served as a central tool for comparing the units generated using ChatGPT. It also formed the basis for the comparison between the AI-generated units and the official reference unit extracted from the tenth-grade physics textbook.

For this study, the analysis was focused on the explicit systematic identification and coding of all learning objectives that were generated from the AI-generated instructional modules, reformulated or newly developed. These objectives were subsequently categorized in accordance with Bloom's Taxonomy and in comparison with the government-stated goals from the guide for tenth-grade physics. Bloom's Taxonomy was selected as the sole analytical framework because it is the most widely recognized and applied framework among teachers and within the Palestinian educational system. It is the official framework adopted in the design of the Palestinian curriculum and in teacher training programs. Consequently, teachers are highly familiar with Bloom's categories and have extensive practical experience in applying them across various instructional elements. For this reason, Bloom's Taxonomy provided the most relevant and contextually appropriate lens for analyzing AI-generated content in this study.

This process used a specifically developed coding table, aligned to each of Bloom's Taxonomy levels. The table included analytical categories and particular indicators permitting the organization and classification of data, enabling identification of recurring patterns that facilitated the development of a particular interpretive understanding.

Additionally, a cross-case analysis was conducted among the AI-generated units themselves to identify shared features and differences resulting from the teachers' backgrounds and their strategies in directing the AI during the design process. To ensure reliability, the analysis was also conducted by another researcher, and the inter-coder agreement was calculated using Holsti's formula, yielding an agreement rate of 91 %.

3.6. Analysis method for semi-structured interviews and the interaction monitoring template

The study employed Braun and Clarke's (2006) model for thematic analysis to examine teachers' responses to the semi-structured interview questions as well as the data recorded in the interaction monitoring template between the teacher and ChatGPT. The analysis began with transcribing the interviews, followed by thorough reading of the transcripts to gain a comprehensive understanding. The data were then coded, with each meaningful expression assigned a label that reflected its implicit meaning.

Related codes were grouped and reorganized into main themes that reflected the overarching dimensions of the data. The researcher

Table 1

Distribution of the formal Unit's Objectives According to Bloom's Taxonomy of Cognitive Objectives.

Lesson	Remembering	Understanding	Applying	Analyzing	Creating	Evaluating	Total
1. Physics and Measurement	19	11	4	1	0	1	36
2. Vectors	5	4	9	0	1	2	21
3. Describing Motion	8	10	8	4	0	0	30
4. Newton's Laws	19	14	13	6	2	0	54
total	51	39	34	11	3	3	141
Percentage	36 %	28 %	24 %	8 %	2 %	2 %	100 %

carefully reviewed these themes to ensure they were comprehensive and non-overlapping. Each theme was supported with direct quotations from the interview transcripts to enhance interpretive credibility.

To strengthen the credibility of the qualitative analysis, the researcher implemented several procedures, including returning to the original texts to minimize interpretive bias, comparing categories to ensure internal and external consistency, and conducting double-checks of selected segments with a co-researcher to establish inter-rater reliability. Additionally, member checking was carried out by sharing analytical summaries with the participants to confirm the accuracy of representing their views. A detailed audit trail was also maintained, documenting all stages of the analysis to enhance transparency and traceability of the research.

To further ensure internal validity and verification of findings, a triangulation process was conducted by comparing the interview data with the results of the content analysis of the instructional units and the data derived from the interaction monitoring template. This triangulation helped assess the alignment between what teachers expressed verbally and what they actually produced, enabling a deeper and more robust interpretation.

All phases of the analysis were carried out manually, without the use of specialized qualitative data analysis software, due to the relatively limited scope of data and the emphasis on close engagement with precise educational texts. This manual approach allowed for a deeper interpretive understanding that was not constrained by algorithmic structures. An inductive approach was intentionally adopted to avoid imposing predefined analytical models on the data and to allow thematic patterns to emerge organically from within the experience itself—reflecting the exploratory nature of the study, grounded in detailed observation and interpretive reading of meaning.

The findings from this analysis will be presented in the results chapter under clearly defined thematic categories, supported by examples from the original interview transcripts or excerpts from the unit content, along with descriptive tables that illustrate the distribution of themes and indicators across the different cases.

3.7. Data collection

The study employed an integrated design based on **methodological triangulation** to enhance the credibility and interpretive coherence of the findings. Three primary tools were utilized, each assigned specific functional and temporal roles across the different stages of the research to contribute to a comprehensive understanding of the phenomenon under investigation. These tools included: (1) content analysis of instructional units generated using artificial intelligence, (2) semi-structured interviews with teachers, and (3) the interaction monitoring template documenting teacher engagement with ChatGPT.

The first step involved extracting and analyzing the content of the “Mechanics” unit from the teacher’s guide of the Palestinian tenth-grade physics curriculum. This analysis included specific behavioral objectives for each lesson, distributed across the six cognitive levels of Bloom’s Taxonomy. It served as the analytical database for comparison with the learning units created using ChatGPT. The results of this official analysis revealed that the objectives in each lesson predominantly targeted lower-order cognitive skills (remembering, understanding, and applying). This distribution indicates that the structure of the official unit is foundational in nature, focusing more on building students’ basic knowledge rather than fostering higher-order thinking skills.

In the second stage, the participating teachers were provided with the educational objectives derived from the official analysis of the instructional unit (as outlined in the teacher’s guide for tenth-grade physics). They were then asked to design instructional units using ChatGPT and submit them to the researcher once they were satisfied with the outcome. In addition, they were asked to submit their completed interaction monitoring templates documenting their engagement with ChatGPT during the design process.

In the third stage, semi-structured interviews were conducted with each teacher. These interviews aimed to explore their perceptions regarding the use of ChatGPT in designing instructional units.

3.8. Validity, reliability, and ethical considerations

Validity and reliability were established through the adoption of methodological triangulation, whereby data were collected using three different tools. This diversity in data collection allowed for the phenomenon to be examined from multiple perspectives and contributed to the development of a coherent interpretive understanding.

To enhance the validity of the data obtained from the interviews, the researcher employed member checking by sharing the findings with participants for their review and feedback. This process aimed to ensure the accurate representation of their statements and practices, reduce interpretive bias, and strengthen trust between the researcher and the participants.

To reinforce construct validity during the development of analytical tools, the researcher consistently referred back to the study’s objectives when defining the analytical categories and designing the coding tables. This ensured methodological coherence and prevented fragmented or incidental analysis. In order to verify the face validity of the indicators used to represent the cognitive levels in Bloom’s Taxonomy, the researcher presented the indicators to a panel of five experts—including both practicing teachers and experienced instructional designers—who confirmed the indicators’ appropriateness in accurately reflecting the levels of Bloom’s Taxonomy.

For ensuring internal reliability, the researcher invited an external specialist to review selected excerpts from the interview transcripts and unit analyses, covering approximately 30 % of the analyzed data. The level of agreement between the researcher’s coding and that of the external specialist was verified, which added greater credibility to the study’s findings. Also the same analytical tool used for evaluating the instructional units designed by the teachers was applied to each unit independently, using standardized coding procedures, which enhanced the reliability and traceability of the comparative results.

In terms of ethical compliance, participants were informed by the researcher that participation in this study was entirely voluntary, and that they had the right to decline or withdraw at any stage of the process without any obligation. Written informed consent was obtained from all participants. The researcher committed to maintaining strict confidentiality of any data that might potentially reveal their identities. To further protect anonymity, pseudonyms were used to refer to the teachers; the first, second, and third participants were identified using numerical codes (1, 2, and 3) throughout all stages of analysis and reporting.

The ethical commitment extended to all phases of data handling, including the instructional units designed using ChatGPT, the audio recordings of the semi-structured interviews, and the transcriptions of teacher interactions with ChatGPT. All data were stored securely and used solely for the purposes of this research. A detailed audit trail was maintained, documenting each step of the analysis process—from initial coding to the development of final themes—thus enabling future review or verification by independent researchers.

Finally, this study does not aim to generalize its findings in a statistical sense. Rather, it seeks to provide an in-depth interpretive understanding of teacher behavior and instructional design practices in a real-world setting when using generative AI tools, thereby paving the way for similar studies in other educational contexts.

4. Results

4.1. Research question one

The first research question stated: *What is the distribution of the learning objectives included in the units designed using ChatGPT according to*

Bloom's cognitive taxonomy?

To answer this question, the instructional units designed by the teachers using ChatGPT were analyzed in accordance with Bloom's taxonomy of cognitive objectives (Remembering, Understanding, Applying, Analyzing, Creating, and Evaluating). The analysis aimed to identify both the quantity and nature of the objectives embedded in each unit. The content analysis revealed a clear progression in cognitive levels, although variations were observed in the representation and distribution of these levels across the different units.

The unit designed by Teacher 1—the least experienced among the participating teachers—included a total of 126 learning objectives, distributed across Bloom's levels as follows: Remembering (65 objectives, 52 %), Understanding (28 objectives, 22 %), and Applying (33 objectives, 26 %). The unit did not contain any objectives related to higher-order thinking skills (Analyzing, Creating, or Evaluating). This distribution clearly indicates a strong emphasis on lower-order cognitive levels (Remembering, Understanding, and Applying). Table (2) presents these findings in detail.

Meanwhile, the unit designed by Teacher 2 (with intermediate experience) included 151 learning objectives, distributed across Bloom's cognitive levels as follows: Remembering (50 objectives, 33 %), Understanding (44 objectives, 29 %), Applying (41 objectives, 27 %), Analyzing (6 objectives, 4 %), Creating (7 objectives, 5 %), and Evaluating (3 objectives, 2 %). A clear representation of the lower-order cognitive levels (Remembering, Understanding, Applying) is observed, with a slight emphasis on the application level. There is also a minimal, though present, representation of higher-order cognitive levels (Analyzing, Creating, Evaluating), which remains relatively low. This unit shows an increase in the total number of learning objectives compared to the unit designed by the novice teacher, and demonstrates greater variety, as it includes objectives across all six levels of Bloom's taxonomy. However, the distribution of higher-order objectives (Analyzing, Creating, Evaluating) remains limited. Table 3 presents these results in detail.

Meanwhile, the unit designed by Teacher 3 (the most experienced) included 146 learning objectives, distributed across Bloom's cognitive levels as follows: Remembering (30 objectives, 21 %), Understanding (51 objectives, 35 %), Applying (51 objectives, 35 %), Analyzing (8 objectives, 5 %), Creating (3 objectives, 2 %), and Evaluating (3 objectives, 2 %). There is a clear representation of lower-order cognitive levels (Remembering, Understanding, Applying), with an equal distribution between Understanding and Applying. Although limited, higher-order cognitive skills (Analyzing, Creating, and Evaluating) are also present.

This unit contains a greater number of objectives compared to the unit designed by the novice teacher and slightly fewer than the one designed by the intermediate teacher. It also demonstrates diversity in cognitive levels, covering all six levels of Bloom's taxonomy, although the higher-order levels remain underrepresented. Table 4 presents these results in detail (see Table 5).

Table (5) and figure (1) presents a comparative summary of the percentage distribution of learning objectives across Bloom's Taxonomy levels for the three teachers' units, highlighting the differences in their use of lower- and higher-order cognitive skills.

To illustrate differences in formulation and cognitive depth, Table 6

presents sample objectives from the official unit and the AI-generated units designed by the three teachers, organized by Bloom's Taxonomy levels.

This comparison shows that while the official objectives provided to ChatGPT were generally preserved, additional objectives emerged during the teachers' interaction with the tool. These new objectives, particularly at the higher Bloom's levels, were not part of the official unit, indicating that adherence to the formal objectives was not strict. Instead, the open-ended interaction between the teachers and ChatGPT led to the generation of objectives beyond the official curriculum.

4.2. Research question 2

The second research question stated: *What are the similarities and differences between the learning objectives in the units designed using ChatGPT and those in the formal unit, in terms of cognitive depth according to Bloom's Taxonomy?*

This study explored the extent of divergence or alignment between the learning objectives included in the instructional units designed by the participating teachers using ChatGPT and the objectives outlined in the official instructional unit from the tenth-grade physics textbook. The analysis was conducted based on Bloom's taxonomy of cognitive objectives. The official analysis provided in the teacher's guide for the tenth-grade physics curriculum served as the benchmark reference, representing the officially approved standard in the Palestinian educational context.

The analysis showed that the official unit included 141 learning objectives, most of which were concentrated at the lower levels of Bloom's taxonomy (Remembering and Understanding), accounting for approximately 64 % of the total objectives. The higher-order cognitive levels (Analyzing, Creating, Evaluating) were only symbolically represented, reflecting a traditional instructional design approach that focuses more on factual knowledge and interpretation than on the development of higher-order thinking skills.

In contrast, the unit designed by Teacher 1 (novice) included fewer objectives compared to the official unit. The analysis revealed an unbalanced distribution of objectives, with all of them focused exclusively on lower-order cognitive levels (Remembering, Understanding, and Applying). Nearly half of the objectives fell under the "Remembering" level, with the rest divided between "Understanding" and "Applying."

The unit designed by Teacher 2 (intermediate experience) showed a different pattern compared to that of Teacher 1. In terms of level diversity, the objectives were distributed across all six levels of Bloom's taxonomy, including minimal representation of higher-order thinking skills (Analyzing, Creating, Evaluating). Although lower-order levels still dominated the distribution, the inclusion of higher-level objectives—even if limited—indicated a broader cognitive scope. The unit included 151 objectives, which is more than the 126 objectives in Teacher 1's unit and 10 more than the official unit. Compared to the official unit, Teacher 2's unit had both a greater number of objectives and broader representation across Bloom's levels—though, again, lower levels were more heavily represented.

The unit designed by Teacher 3 (expert) also revealed a distinct pattern when compared to the units created by Teachers 1 and 2. Like Teacher 2's unit, it included objectives across all cognitive levels, with

Table (2)

Distribution of Objectives in the Instructional Unit Designed Using ChatGPT by the teacher 1 (Novice Teacher).

Lesson	Remembering	Understanding	Applying	Analyzing	Creating	Evaluating	Total
1. Physics and Measurement	19	9	6	0	0	0	34
2. Vectors	12	2	8	0	0	0	22
3. Describing Motion	10	7	9	0	0	0	26
4. Newton's Laws	24	10	10	0	0	0	44
total	65	28	33	0	0	0	126
Percentage	52 %	22 %	26 %	0 %	0 %	0 %	100 %

Table 3

Distribution of Objectives in the Instructional Unit Designed Using ChatGPT by the teacher 2 (intermediate experience Teacher).

Lesson	Remembering	Understanding	Applying	Analyzing	Creating	Evaluating	Total
1. Physics and Measurement	15	14	9	2	2	1	43
2. Vectors	13	7	7	1	1	1	30
3. Describing Motion	8	10	13	1	2	1	35
4. Newton's Laws	14	13	12	2	2	0	43
total	50	44	41	6	7	3	151
Percentage	33 %	29 %	27 %	4 %	5 %	2 %	100 %

Table 4

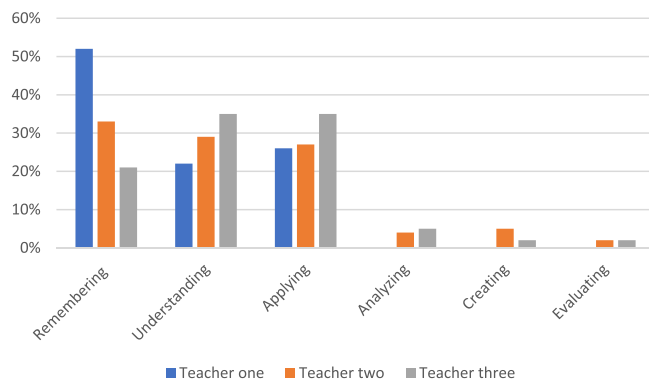
Distribution of Objectives in the Instructional Unit Designed Using ChatGPT by the teacher 3 (experience Teacher).

Lesson	Remembering	Understanding	Applying	Analyzing	Creating	Evaluating	Total
1. Physics and Measurement	8	10	15	2	0	1	36
2. Vectors	4	13	12	1	1	0	31
3. Describing Motion	8	12	13	2	1	1	37
4. Newton's Laws	10	16	11	3	1	1	42
total	30	51	51	8	3	3	146
Percentage	21 %	35 %	35 %	5 %	2 %	2 %	100 %

Table 5

Percentage distribution of learning objectives across Bloom's Taxonomy levels in the units designed by the three teachers.

Teacher	Percentage					
	Remembering	Understanding	Applying	Analyzing	Creating	Evaluating
1	52 %	22 %	26 %	0 %	0 %	0 %
2	33 %	29 %	27 %	4 %	5 %	2 %
3	21 %	35 %	35 %	5 %	2 %	2 %

**Fig. 1.** Percentage distribution of Bloom's Taxonomy levels in the AI-generated units.

limited but present representation of higher-order levels (Analyzing, Creating, Evaluating). Compared to Teacher 1's unit, it included more objectives; compared to Teacher 2's, slightly fewer; and slightly more than the official unit. In terms of distribution, the objectives covered all levels of Bloom's taxonomy with more balanced representation than in Teacher 1's and the official unit. Its distribution was also somewhat similar to that of Teacher 2's unit.

Table 6

Sample comparison of learning objectives across Bloom's levels (Official vs. AI-generated).

Bloom's Level	Official Unit Objective	Teacher 1 (Novice)	Teacher 2 (Intermediate)	Teacher 3 (Expert)
Remembering	Define the concept of vector.	Define the concept of vector.	Define the concept of vector.	Define the concept of vector.
Understanding	Explain Newton's First Law with examples.	Explain Newton's First Law with examples.	Explain Newton's First Law with examples.	Explain Newton's First Law with examples.
Applying	Solve problems using Newton's Second Law.	Solve problems using Newton's Second Law.	Solve problems using Newton's Second Law.	Solve problems using Newton's Second Law.
Analyzing	Distinguish between balanced and unbalanced forces.	— (Not present)	Analyze the forces acting on a moving vehicle.	Analyze the effect of friction on motion in daily-life contexts.
Creating	Design a simple experiment to measure acceleration.	— (Not present)	Design a classroom activity to demonstrate acceleration.	Design a local-material experiment to calculate acceleration.
Evaluating	Evaluate the accuracy of a measurement.	— (Not present)	Evaluate the reliability of an experiment setup.	Evaluate the limitations of Newton's Laws in real-world contexts.

4.3. Research question 3

The third research question stated: *How did the teachers describe their experience using ChatGPT to design an instructional unit, in terms of interaction with the tool, challenges encountered, and their evaluation of the resulting output?*

To address this question, the researcher analyzed the responses of the three teachers to the semi-structured interview questions, through which they shared their experience of designing the instructional unit using ChatGPT, along with their responses to the interaction monitoring template documenting their engagement with the tool. These responses were categorized into five main themes: positive interaction with ChatGPT, the need for repeated guidance of the tool, skepticism regarding the quality of ChatGPT's designs, ChatGPT's control over the conversation, and overall positive evaluation of the experience. A detailed presentation of each theme follows.

1. Positive Interaction with ChatGPT

All three teachers expressed amazement at ChatGPT's capabilities in generating content and praised the tool's speed in doing so. Teachers 2 and 3 described the experience as "enjoyable" and "inspiring," reflecting a positive psychological and intellectual openness to using the tool in future instructional design endeavors. Despite the varying levels of teaching experience among participants, this positive impression was consistently expressed in different forms, suggesting that the technological capabilities of the tool played a motivating role regardless of the users' background.

Teacher 3 commented: *"I feel like my profession is under threat ... It can do what an entire team does."*

Teacher 2 said: *"I enjoyed designing the unit ... It motivated me to use it again in the future."*

Teacher 1 remarked: *"The program is amazing; it has tremendous potential."*

These positive impressions were not only evident in the interviews but were also reinforced through the responses in the interaction monitoring templates. Teacher 3, for instance, expressed his admiration from the outset and wrote in the section evaluating the output:

"The results surprised me from the very first attempt. I used what ChatGPT provided as a foundation and then restructured it to make it more suitable for my students."

Similarly, Teacher 2 indicated flexibility in his use of the tool, stating:

"The generated questions were excellent ... They made me reconsider how I pose questions in class."

Teacher 1 wrote:

"The idea that the tool provides a ready-made plan so quickly was very helpful."

This positive interaction was reflected in the quality of the learning objectives generated in the designed units. The analysis showed that the teachers who expressed enthusiasm during their interaction were able to guide the tool effectively toward generating diverse and well-structured learning objectives aligned with Bloom's taxonomy. By contrast, the unit produced by Teacher 1 appeared structurally limited and largely reflected a simple reproduction of the formal objectives. In practice, this meant that the unit was poor in terms of content and activities, resembling technical responses to some of the official objectives while neglecting others. The generated text lacked curriculum aesthetics, as it was presented in a dry, mechanical style that did not provide the kind of engaging discourse capable of bringing the scientific concepts closer to students or motivating them to engage with the material.

This convergence between interview responses and content analysis reinforces the credibility of this theme and highlights a strong relationship between openness to the tool and the quality of pedagogical interaction with it. The positive impressions expressed by Teachers 2 and 3 corresponded with the variety and structure of the learning objectives in their units. Their enthusiasm appears to have enabled them to direct the tool toward producing content across a broader range of cognitive levels.

However, the positive impression expressed by Teacher 1 did not align with the actual outcomes of his designed unit. The number of objectives generated was lower than the number of objectives provided to him, which may indicate the influence of pedagogical experience in designing effective instructional units using ChatGPT.

2. The necessity of repeatedly guiding the tool

All three teachers, despite expressing positive impressions of ChatGPT's capabilities during the interviews, emphasized that achieving a satisfactory output from the tool required repeated guidance and precise instructions. They noted the need to reformulate or elaborate on their prompts multiple times, indicating that the interaction with the tool was not automatic as it did not respond exactly as intended from the outset. This was evident in their statements:

- Teacher 1: *"I had to write the objectives in a Word file and send them to it, it did not respond when I fed it as images. But once I fed the objectives into ChatGPT, completing the task didn't take much time."*
- Teacher 2: *"I thought I would get the design in record time, but I had to give ChatGPT a lot of instructions to get a good unit."*
- Teacher 3: *"I didn't like its initial response when I gave it the request. So I gave more instructions, and honestly, it didn't respond the way I expected. It needs many directions to give you what you want."*

These statements were directly supported by the data from the interaction monitoring templates, which revealed noticeable differences between teachers in the number of instructions provided to the tool:

- Teacher 1 submitted only 12 instructions, most at the beginning of the conversation, and then relied on the outputs without further modification.
- Teacher 2 submitted 22 instructions, including several rephrasings of the prompts and elaborations on instructional activities.
- Teacher 3 submitted 29 instructions, employing a task segmentation approach and revising the outputs based on contextual needs.

These variations in the level of guidance given to ChatGPT during the unit design process were reflected in the quality of the learning objectives produced. Teacher 3 demonstrated greater ability to achieve a balanced distribution across Bloom's taxonomy compared to Teacher 1, whose objectives were limited to the lower levels (Remembering and Understanding). This illustrates that ChatGPT does not automatically generate high-quality content simply upon request—it requires skilled prompting and pedagogical awareness to translate user commands into precise and effective educational instructions.

This theme clearly shows that repeated guidance is not merely a reflection of the tool's limitations, but rather a core component of successful instructional design using artificial intelligence. A teacher's ability to refine this interaction directly impacts the quality of the educational output generated.

3. Skepticism Toward the Design Provided by ChatGPT:

Teachers' 3 and 2 responses during the semi-structured interviews revealed clear signs of doubt regarding ChatGPT's ability to accurately reflect instructional objectives. This skepticism was evident in their responses to the interview questions. For example, Teacher 3 indicated

during the interview that he was not confident the tool had understood his intent properly, which led him to repeat his instructions multiple times. He remarked:

"I had to repeat the instructions several times ... I wasn't sure it understood the purpose behind my request."

Similarly, Teacher 2 expressed reservations about relying solely on the initial outputs, stating:

"ChatGPT provides good content ... but the amount of prompting it requires makes you question the quality of its output, even if the final result is acceptable."

These responses reflect a cautious stance from the teachers toward the tool's results—even when accurate and structured inputs, such as officially classified objectives, were provided.

This skepticism was further supported by the interaction monitoring templates. In the section on challenges encountered, teachers wrote the following:

- Teacher 3 noted under the question about ambiguity, repetition, or misalignment in the outputs:

"The fact that ChatGPT did not respond correctly the first time made me doubt its ability to reflect all the objectives I provided for the instructional unit."

- Teacher 2, in response to whether he would consider using the AI-designed unit in his teaching, wrote:

"It requires a thorough review of the designed unit and careful comparison with the original unit in the textbook."

- Teacher 1, however, offered a contrasting opinion. When asked the same question, he stated:

"I can use the AI-designed unit because it gives me more freedom to work, rather than being restricted by the official textbook unit."

These cautious attitudes aligned with the results of the content analysis. Teachers 2 and 3, who expressed less confidence in the AI-generated output, were the ones who actually revised and restructured parts of the units. This was reflected in the closer alignment between the number and distribution of objectives in their units and those found in the official unit, as categorized according to Bloom's taxonomy.

4. ChatGPT's Control Over the Dialogue:

Teachers' responses revealed an implicit sense that ChatGPT exerted significant influence over the direction of the dialogue during the design process, with the interaction not always being balanced or fully controlled by the user. In their interviews, some teachers indicated that they found themselves following ChatGPT's suggestions or going along with its direction, even when that wasn't their original intention. This was reflected in their comments:

- Teacher 1: *"I think one of the challenges I faced was the tool's openness. It kept asking me whether I wanted to adjust or add things, and I felt it was guiding me, not the other way around."*
- Teacher 2: *"Sometimes it leads you, and if you're not paying attention, you find yourself changing your original idea just to go along with its direction."*

It was also evident that Teacher 1 showed a greater degree of compliance, likely due to overconfidence in ChatGPT's output and the

limited number of instructions he provided.

In the interaction monitoring template, Teacher 2 noted under "Challenges":

"Sometimes ChatGPT went into unnecessary detail, and I just followed along until I realized I had ended up somewhere else."

Similarly, Teacher 3 wrote under "Interaction Style":

"It needs a user who knows exactly what they want, or else you'll get lost."

These reflections suggest that ChatGPT may exert a form of "interactive dominance" over the course of the dialogue, where the teacher shifts from being a director to a recipient of the tool's suggestions. This dynamic may affect the originality of the instructional design, or lead to partial divergence from the intended learning objectives.

This dominance was also reflected in some of the units designed using ChatGPT. In certain cases, instructional plans exhibited a general, formulaic structure that resembled the typical output pattern of the tool, rather than aligning closely with the teacher's intended objectives—as was the case with Teacher 1. This implies that the tool's control over the interaction may go beyond technical influence and extend to shaping pedagogical decisions made by the teacher during the design process.

5. Positive Evaluation of the Experience:

Despite some reservations expressed during different stages of interaction, all three teachers agreed that using ChatGPT to design an instructional unit was a valuable and engaging experience. In the interviews, Teacher 3 described the experience as encouraging, saying: *"I enjoyed designing the unit ... it motivated me to use it again in the future."*

Teacher 2 stated plainly: *"This was a great experience—it opened new horizons for me."*

Teacher 1 similarly shared: *"I'm grateful for this experience."*

This sentiment was also echoed in the interaction monitoring templates. Teacher 1 wrote in the final evaluation section:

"This experience gave me an idea I had never considered—an entire unit completed in such a short time."

Teacher 2 noted that the experience increased his awareness of new tools, writing:

"Even if I don't use it exactly as it is, the experience helped me think differently."

Teacher 3 wrote:

"I will definitely use ChatGPT more in my professional work."

These reflections indicate that, despite the challenges, interacting with ChatGPT contributed to the teachers' pedagogical and conceptual development. It helped shape new perspectives on instructional design and highlighted the potential of technological tools to support classroom practices, particularly in contexts where teachers face limitations in time and resources.

5. Discussion

This discussion aims to interpret the findings derived from analyzing the educational objectives in the units designed using ChatGPT, comparing them to the formal objectives found in the original curriculum unit, and analyzing teachers' experiences in interacting with the tool. This section addresses the study's three research questions through a pedagogical reflection based on practical realities and informed by relevant literature.

5.1. First: discussion of the first question

What is the distribution of educational objectives in the units designed using ChatGPT according to Bloom's Taxonomy?

The analysis of the three educational units designed by the participating teachers using ChatGPT revealed clear variation in the representation of cognitive levels according to Bloom's Taxonomy. Most of the objectives in the unit designed by the novice teacher were limited to the lower levels (remembering, understanding, applying), with no representation of the higher-order levels (analyzing, evaluating, creating). The unit designed by the second teacher included objectives distributed across all levels, although with varying proportions and limited representation of higher levels. The third teacher's (expert) unit showed a relatively balanced distribution, with a stronger representation of higher-level objectives compared to the other two.

The analysis of these units indicates a clear disparity in how well the objectives reflect the levels in Bloom's Taxonomy. All of the objectives in the first teacher's unit were situated within the lower cognitive levels. The second teacher's unit contained a greater number and variety of objectives, although still with limited higher-level representation. The third teacher's unit featured a more balanced distribution and included objectives from all cognitive levels, though not equally represented. This variation may be attributed to the teacher's level of experience, as evidenced by the complete absence of higher-order thinking objectives in the novice teacher's unit and their partial presence in the more experienced teachers' units.

These findings are consistent with (Ullmann et al., 2024), who observed that shallow AI-generated knowledge tends to emerge when there is minimal or unskilled human intervention.

5.2. Second: discussion of the second question

What are the similarities and differences between the educational objectives in the units designed using ChatGPT and the official unit in terms of cognitive depth according to Bloom's Taxonomy?

The formal analysis of the "Mechanics" unit in the tenth-grade physics curriculum (Palestinian Ministry of Education, 2019) shows a significant emphasis on lower-order cognitive skills, with minimal representation of higher-order thinking skills. In contrast, the units designed using ChatGPT by teachers 2 and 3 (both considered experienced teachers) included objectives spanning the higher levels of Bloom's taxonomy.

This divergence indicates that generative AI tools, while inherently automatic, are capable of producing designs rich in diversity and cognitive depth—though this does not happen automatically. Such outcomes require intentional, skilled guidance from teachers. One possible explanation is that novice teachers may prioritize output quantity over quality and place excessive trust in technological tools, whereas experienced teachers leverage their professional expertise to provide nuanced prompts that yield deeper instructional content.

These findings align with Ullmann et al. (2024), who emphasized that AI can produce effective instructional content when used as a companion to professional human expertise rather than a replacement.

5.3. Third: discussion of the third question

How did teachers describe their experience using ChatGPT to design an instructional unit in terms of interaction with the tool, challenges, and evaluation of the output?

The analysis of interviews and interaction logs with ChatGPT revealed five key themes: positive interaction with the tool, the need for repeated prompting, skepticism about the quality of outputs, ChatGPT's control over the dialogue, and an overall positive evaluation of the experience.

First, all teachers initially expressed positive interactions and praised ChatGPT's ability to generate content quickly and flexibly. This aligns

with findings by Castillo et al. (2024), who noted that AI use in education can stimulate innovative pedagogical thinking.

Second, teachers reported needing to issue repeated prompts to obtain satisfactory results, which supports Cain's (2024) insights on the importance of prompt engineering in improving the accuracy of AI-generated outputs.

Third, some teachers expressed doubt regarding ChatGPT's ability to accurately interpret their instructional intentions, leading to a reluctance to accept outputs as-is. These findings echo the concerns raised in studies by Alkaissi and McFarlane (2023) and Adeshola and Adepoju (2023).

Fourth, several teachers felt that ChatGPT directed the conversation, which at times compromised their autonomy in making instructional decisions. This concern was previously addressed by Garrison and Kanuka (2004), who highlighted the risk of technological dominance in educational processes.

Finally, despite these reservations, all three teachers expressed general satisfaction with the experience, acknowledging that it opened new possibilities for instructional design. This conclusion aligns with similar findings in Kayabaş (2024), who emphasized the reflective and developmental potential of engaging with generative AI tools in education.

6. Conclusion

This study highlights the growing potential of generative artificial intelligence tools—particularly ChatGPT—in supporting instructional design processes within school-based educational environments. Its findings indicate that the effectiveness of such tools greatly depends on how well they are guided by the teacher. The study shows that this guidance is closely linked to the teacher's professional experience, which plays a significant role in harnessing the latent power of generative AI. The instructional units designed by experienced teachers demonstrated a higher ability to direct AI tools toward generating diverse and balanced learning objectives that reflect the levels of Bloom's Taxonomy.

The study also reveals that interacting with ChatGPT is not a linear or automated process; rather, it requires critical reflection and continuous revision of its outputs. Effective use of the tool is inseparable from the educational context in which it is applied. Thus, the study affirms that artificial intelligence should not be viewed as a replacement for the instructional designer, but rather as a supportive partner that can enhance the teacher's instructional creativity and efficiency—when used within an informed and structured pedagogical framework.

Accordingly, the study calls for a re-evaluation of the relationship between technology and educational practice, where teachers are placed at the center of guiding the use of AI tools in education, without compromising the cognitive and pedagogical standards that ensure content quality and meaningful learning experiences.

Moreover, the study emphasizes that interaction with ChatGPT is not automatic but instead requires ongoing reflection and critical assessment of the generated texts. For the tool's use to be effective, it must be integrated into a well-defined educational context, where the features of that context serve as essential inputs for the application.

In light of these conclusions, the study recommends raising awareness of the relationship between ChatGPT and instructional design. It further calls for training teachers to use AI within professional frameworks to optimize its educational value and for developing practical guides that assist teachers in effectively using the tool in instructional design contexts. In addition, the study recommends developing pre-configured prompts explicitly aligned with Bloom's Taxonomy to serve as scaffolding for novice teachers, helping them to generate more cognitively balanced instructional units.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

Declaration of the use of AI assisted technologies

During the preparation of this work, the author used ChatGPT for proofreading purposes. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendices (1). coding example

Script	Theme
T3. The numerous instructions you need to provide for it to understand what you want made me unconvinced that ChatGPT can generate a dependable instructional unit. I believe that a significant portion of design time must be allocated to deciding whether the generated text actually matches my intentions. Sometimes, I requested a specific task, such as a cooperative interactive activity for conducting an experiment, but instead received something superficial, forcing me to repeatedly clarify my instructions. I would often smile and question why I need this application at all if I can accomplish the task myself. Honestly, it requires considerable time and multiple instructions to produce what you desire; it rarely understands you the first time. Because of this, it took me some effort to design the unit using ChatGPT. I think ChatGPT is a useful tool when placed in the hands of someone who is aware and knows exactly what they want, but it might not be suitable for someone merely looking for quick text generation to save time. If you're not careful, you'll find it leading you far from what you initially wanted, because, for instance, it suggests something that you like, but you're not satisfied with its execution. You ask it to improve, and it responds, and so forth, until you eventually find yourself forgetting what you originally requested. Sometimes ChatGPT went into unnecessary detail, and I just followed along until I realized I had ended up somewhere else	Skepticism Toward the Design Provided by ChatGPT The necessity of repeatedly guiding the tool ChatGPT's Control Over the Dialogue

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agencies in the public, commercial, or not-for-profit sectors.

Ethics approval statement

The study was conducted in accordance with the Declaration of Helsinki. Ethical approval was obtained from the Institutional Review Board (IRB) at An-Najah National University (IRB Protocol Number: Hum.ESci. May. 2025/26). Informed consent was obtained from all participants prior to their inclusion in the study.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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